

FraudGuard AI: Architectural & Forensic Review (SIH 26106)

Feature	SIH Requirement	Current Implementation	Accuracy	Weakness / Recommended Fix
1. Email header forensics	Extract metadata transparently.	Extracts From, To, Subj, Auth. Hashes via SHA-256.	High	Misses X-Mailer, Message-ID. Fix: Extract in parser.py.
2. Received hop reconstruction	Chronological hop chain.	Regex on Received headers & reverses them.	Medium/High	Complex headers might fail. Fix: Use mail-parser lib.
3. Sender spoofing detection	Detect forged senders.	Checks Auth-Results (Alignment).	High	Misses homoglyphs. Fix: Add Levenshtein distance check.
4. SPF/DKIM/DMARC	Strict auth checks.	Parses Auth-Results string for PASS/FAIL.	Medium	Relies on upstream MTA. Fix: Add dkimpy to manual verify.
5. Domain alignment	Check Return-Path vs From.	Strict string comparison.	High	Misses org domain (mail.domain.com). Fix: Use tldextract.
6. IOC extraction	Extract Domains, IPs, URLs.	Extracts IPs, domains, URLs via Regex.	Medium	Misses attachment hashes. Fix: Add email.iter_attachments().
7. IOC classification	Classify (Benign/Malicious).	Lists IOCs without classification.	Low	All look suspicious. Fix: Classify CDNs as BENIGN.
8. URL analysis	Analyze URL risk/redirects.	Extracts raw URL text only.	Low	No reputation check. Fix: Add Google Safe Browsing API.
9. IP intelligence	ASN, ISP, Provider.	Queries ip-api.com.	Medium	Free tier, no reputation. Fix: Add AbuseIPDB/VirusTotal.
10. Geolocation	Accurate physical location.	Uses ip-api.com for City/Country.	Medium	IP geolocation is imprecise. Fix: Keep 'Probable' terminology.
11. VPN/Proxy/Tor detection	Detect obscured infra.	String match on ISP name ('vpn', 'cloud').	Low/Medium	Misses residential proxies. Fix: Add Proxy/Tor exit DB.
12. Threat correlation	Link infrastructure evidence.	Vis.js graphs Email -> IP -> ISP.	High	Doesn't graph URLs. Fix: Add URLs/domains as Vis.js nodes.
13. Attribution limitations	Do not overclaim.	Origin Confidence drops 40% if cloud/VPN.	High	Perfectly aligned with SIH. No fix needed.
14. AI/NLP analysis	Detect social engineering.	TF-IDF + keywords.	Medium	TF-IDF lacks semantics. Fix: Use HuggingFace transformer.
15. Explainable risk scoring	Transparent score.	Aggregates evidence, outputs trust/risk strings.	High	Excellent implementation. No fix needed.
16. Traceability confidence	Rate hop reliability.	Deducts for private IPs, proxies, missing headers.	High	Excellent implementation. No fix needed.
17. Forensic report	Exportable case file.	Generates .txt report with SHA-256.	Medium	Standard .txt. Fix: Export highly formatted PDF.

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18. PII masking	Protect privacy.	UI badge only.	Low	Backend doesn't redact. Fix: Add regex to mask emails in text.
19. Security of uploads	Prevent exploits.	Processed as byte stream. HTML tags stripped.	High	Highly secure. Fix: Add file-size limit to prevent DoS.
20. Fake/hardcoded data	Prohibit fake data.	Removed all hardcoded mock VT results.	High	Relies on live extraction. No fix needed.

Judge's Verdict & Summary

The system successfully transitioned from a conceptual mock-up to a technically viable, forensic prototype. It excels in explainable scoring, infrastructure traceability, and strict adherence to evidence.

The biggest remaining weaknesses that a tough judge might poke holes in are IOC Classification (failing to distinguish a benign CDN URL from a phishing link) and PII Masking (claiming it's active when the backend isn't actually redacting the text).